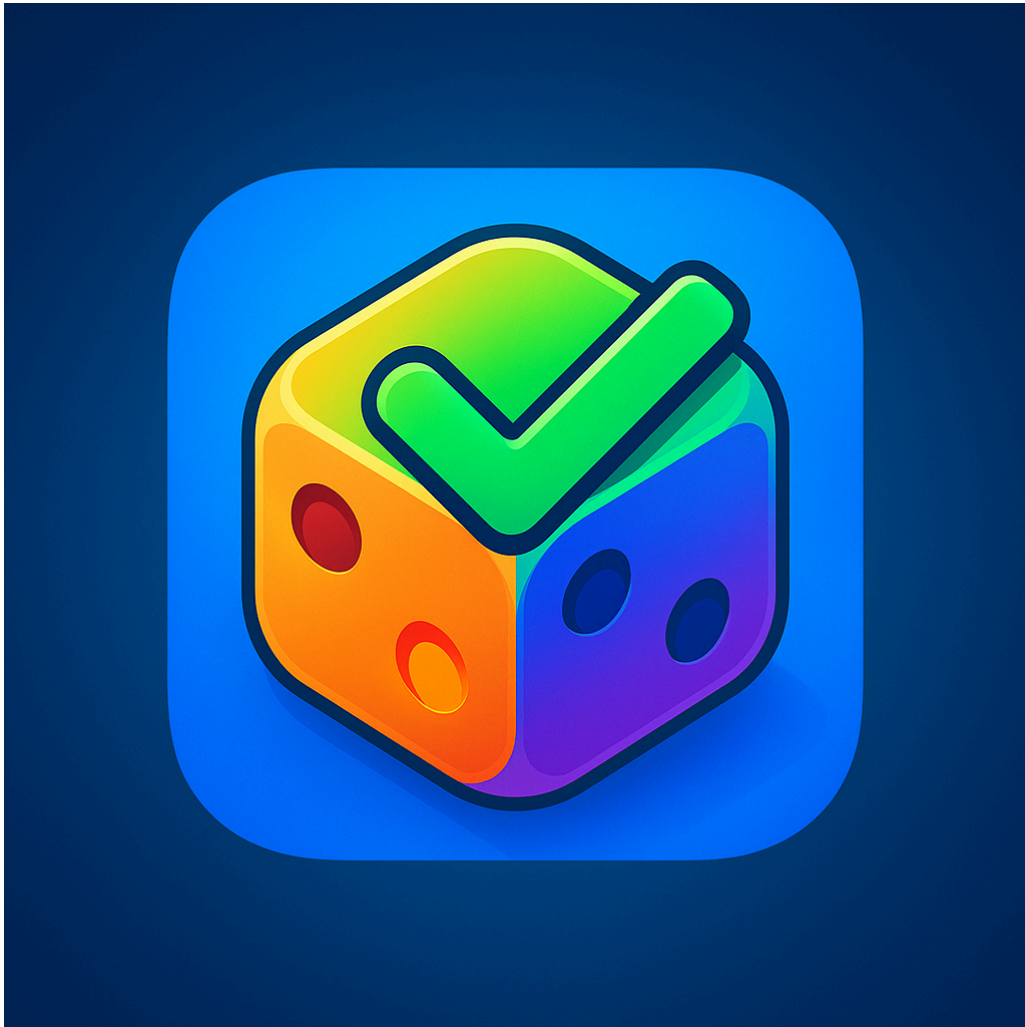


HABITHUB

A multi-generational habit-building and goal-tracking app that combines gamification, family involvement, and social accountability in one transformative hub.



TEAM MEMBERS:

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Course: CSIS 3375 - 001

Requirements Gathering

Purpose or objectives for Requirements Gathering User Study:

- Understanding common habits/traits in children and parents to help focus on our target market for more accuracy in our app development
- Gathering information on parents preferred preferences in terms of how they would like to view family data (using diagrams or just written statements) and also how they would like to control child accounts and privacy methods
- Researching and assessing common gamification methods that children tend to engage in
- Overall understanding on what parents would benefit from using this app and what concepts they would also like to see
- Gathering input on UI design for both target markets (children and parents) to see what their priorities are in terms of app structure and design
- How frequent would they like reminders, notifications to be shown
- Assess different habit hub structures (Classroom, friends, sports team)

Choice of user study and user sample:

User Study: We will be using a dual perspective approach with in-person interviews that are not fully structured, this way we get to have a true open conversation which lets us get deeper in certain topics by asking follow up questions on a specific topic being discussed or briefly go through one topic. By using a semi structured interview it helps us better understand our target markets actual needs as they have more space to deep dive into topics they truly find interesting.

Participants

- Children (ages 6-12) - Children who I (Jayden) currently work with
- Parents - Will include Children's parents (from Jayden's workfield) family members, mentors, high school teachers, and coaches
- Adults (ages 25-50) - additional adults who do not have children but would use our app for personal benefit

Verbal consent will be given before any interview questions are asked, and as well as parental approval before asking any child any interview question.

Draft of the User Study Survey or Interview

Children (ages 6-12)

The structure of these questions are designed to assess the amount of tasks a child has on a daily basis and a deep dive into their creativity and preferences. We decided to shape these questions in a way where we see their imagination in their “perfect” game world. This will help us gamify HabitHub so it can meet the children's preferences. It is focussed less on the functionality of the app but more on the UI design portion.

1. What are some things you do everyday, for example do you make your bed everyday, or brush your teeth every morning and night.
2. Out of the list you just gave me, what do you find the most challenging to do and what do you find the most challenging to remember to do
3. Do you have any tasks that includes the help of others (relatives, family, friends)
4. When you finish your tasks do you get rewarded, and what do you get rewarded with (time on the game, time with toys, stickers, etc)
5. What is your favourite reward you get and why is it your favourite
6. If your tasks could be turned into a video game would that make it more fun for you
7. If the game (app) had a character (yourself) and it would get stronger the more tasks you do would you enjoy that
8. What are some ideas you think you would add to this game

Parents

The structure of these interview questions is designed to see a parents true input on their child's goals and tasks whilst understanding their core ideas on the functionality of the app, such as how they would like to view their data, the app structure and privacy.

1. What are the most important tasks/routines your child does on a daily basis and what habits would you like them to start or improve on.
2. In terms of data, how would you like to view it? Through graphs and visuals or would you like written summaries.
3. What are your main concerns with your child's routines/tasks
4. If possible how would you feel if you could control your child's screen time based on the amount of tasks they have done
5. Is there any additional features you think would be good to add that would benefit you
6. How often would you like to be notified? (notify on every task they have completed or just once they have finished for the day)
7. Would you be okay for gamification in terms of keeping your child engaged and motivated.
8. Would you be okay for to set up goal trackers for your children or give them some freedom to add their own goals as well
9. In terms of your kids' habits, what are some features you'd like to have to help shift their habits into a healthy and positive way.

Adults

Questions are designed for adult personal preference in terms of how they would use the app and for social purposes

1. What are your daily goals and habits
2. What are the goals you find you need most help with to stay consistent
3. What are some habits you are trying to build
4. What motivates you to stay consistent? Would sharing your habits with others to see your progress help?
5. What are your “rewards” when you finish a task or habit? Would rewards such as streaks, badges and visuals make you feel rewarded
6. How often would you like reminders/notifications on tasks and habits you have not done or that are coming up (if you have a specific time you do something)
7. Would you like to share you progress and data with others like friends and family
8. How would you like to see your data; through graphs and visuals or summaries

Data Visualization and Analysis

Sample Size: We have interviewed a total of 17 participants ranging from children to adults.

Data will be collected and organized into a word file summarizing key topics (point form) discussed during interviews and also key works will be taken down this way we can see what our participants are most intrigued in.

We will visualize our data through graphs like bar charts or pie charts. We would use this to compare which features outshines others to ultimately choose how we are going to design our app. For example, we will use a pie chart for certain gamification features. The feature that has the most backing will be the feature we decide to implement in our project. We can also use word clouds to group the keywords we find our participants use. For example, we will use word clouds to see the most common keywords spanned across our target market to see what they all have in common and emphasize on.

Once we have gathered and organized the data we will then proceed to determine and design our app based on the elements that were most desired and talked about in our interviews. For example if the participants all enjoyed the idea of a points reward system we will implement that into our app.

Insights From Results:

Key points discovered amongst participants

Children

- Intrigued with visual rewards, screen time extension, and social interaction between family and friends (fun way to socialize and a sense of competitive ness)
- Habits are still developing so using the app to suggest or help create healthy habits for children
- Children find the gamification very exciting and are very interested
- Most children wanted character customization, and point rewards

Parents

- Majority of parents were enthusiastic about the app as it is a way to get their kids off certain games and focussed on our app instead as it would create healthier habits.
- Simple and easy to read design in terms of data. They would like mainly visual data and some short summaries in terms of assessing data.
- Privacy concerns in terms of sharing child's name and other information. Should only be shown to other family members (have usernames instead of real names for friends)
- Want notifications on all habits/tasks done by their kid

Adults

- Enjoyed the idea of a point system and badge rewards
- Self improvement and consistency were big in conversations with participants
- Visual presentation in terms of progress and data instead of summaries
- Simple and easy to navigate UI
- Adjustable notifications (some would just like to have reminders at the end of the day)

With the data we have collected the users' responses seemed very intrigued with our app. They all believed this would be a good idea and would benefit them and their family. This changes the requirements for our app a little bit as originally we had a standard gamification method (points) but now we would like to gamify our app more as it makes it more engaging and social for all target markets.

REQUIREMENTS GENERATION

Personas and Scenarios:

Persona 1:

Name : Ali

Age : 10

Role : Child

Relation : Child from Jayden's work force

Summary: Ali is a 9 year old kid who is deeply passionate about video games, his favourite games are Roblox and Minecraft. He is also obsessed with sports, specifically soccer. He is a very enthusiastic kid who enjoys trying new things and learning. He tends to have an overall busy schedule as he is in multiple sports, after school programs and classes. With this busy schedule as a kid it is hard to remember to finish all his tasks and would find it beneficial if he would get notified so he remembers to do a task. He is also very intrigued with the gamification and wants to be able to visually see his progress through his character or points.

Goals

- Make routines feel fun and engaging
- Remember to finish all tasks before the day is done
- Wants to explore the gamification material
- Wants to be able to start new routines

Scenario:

After a long day of school, soccer practice and out of school classes, Ali gets home and gets a notification from Habit Hub. He checks HabitHub and sees the routines, and tasks he has (homework, dishes). This helps Ali remember what he has to do before going to bed. After completing his tasks he opens HabitHub and explores the point shop and decides to buy a rare soccer ball accessory for his character.

Personal 2:

Name: Eugene Fung

Role: Parent (2 girls aged 6 and 11)

Relation: Former teacher and mentor for Jayden

Summary:

Eugene is a father of 2 and is mainly concerned on helping them build healthy habits. He prioritizes values and family, he wants his girls to be independent and able to do things on their own even at a young age. For Eugene he is more concerned on the way of life and not necessarily chores around the house. He is looking for consistency with his kids that is maintainable but able to grow. In terms of HabitHub he does not want an overly complicated app he would rather it be straight forward so he can add habits for his kids like brushing teeth every morning and night. He prefers to see his daughters progress through graphs and other visuals. He wants to be able to monitor, create and assign tasks to his daughters.

Goals

- Assign and create tasks to his daughters
- View progress through visuals with some written portions
- Encouraging healthier habits and less of chores
- Focussing on structure and consistency for his kids

Scenario:

Eugene opens HabitHub throughout the day to see what tasks his kids have done so far. He then creates a new task for his kid assigned for Friday night for his girls to get soccer gear and bag ready for Saturday's game. He also creates more routines for 30 minutes of no phone before bed which kids will check off the following morning. At the end of every day Eugene checks his daughters profiles to see how consistent they have been.

Persona 3

Name: Nandani Dhaul

Age: 22

Role: Adult

Relation: Coworker

Scenario:

After working a morning shift Nandani goes to the gym before going to her classes for the day. After she goes to the gym she opens HabitHub to see what stores she will need to go to for her groceries for the week. As she goes store to store she checks them off one by one and then proceeds to go off to her classes for the day. After each class ends she checks them off on HabitHub. Nandani then gets home and cleans up her apartment. At the end of the day she lays in bed looking at HabitHub realizing all the points she has earned for today and decides to go to the point shop to buy some items for her character. Nandani believes HabitHub is a motivator for her to stay productive, consistent and more importantly organized with such a busy schedule.

Requirements Generation

- Data to be visualized mainly with some summaries
- Over-gamification would not hurt
- Simplistic UI for easy access and navigation
- Create, edit, delete tasks/routines
- Interactive dashboard
- Points based on certain tasks (parents can assess points)
- Send reminders and notifications based on users preference
- Include a customizable character
- Daily point shop to encourage rare items (items reset every 24 hours)
- Have lots of cosmetics for your character
- Display rewards and streaks for friends and family to see
- Parent to have access easily to all children individually or as a group
- Strict privacy standard for families
- Social Tab to see how friends and family are doing
- Have multiple hub types such as friends family and teams
- Parents should have full control over child accounts

Synthesis of Requirements Gathering and Generation

High Priority

- Create, edit, and delete tasks/routines (all target markets)
- Visual dashboards
- Simplistic UI
- Customizable characters
- Point system
- Daily point shop
- Reminders and notifications
- Parent dashboard
- Privacy standards
- Easy to read data
- Social tab
- Leaderboard

Medium Priority

- Multiple hub types
- Weekly report
- Monthly report
- Offline functionality
- Trade items from point shop
- Sound effects and animations

Low Priority

- Seasonal items in point shop
- Motivational quotes for adults
- Group challenged based on family structure
- AI integration on habit suggestions

Conclusion

The 3 main concepts taken away from the synthesis of requirements gathering is Gamification, Data visualization, Privacy standards. As a team we will mainly focus on these 3 core concepts to build our application.

AI Use Section

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used
ChatGPT - LOGO	LOGO 4o	Generating Logo (not official logo)
ChatGPT	GPT 5	Asking for ways to visualize data

Value Addition:

For the logo I asked for specific details like more gamification for generating the logo. This log we have used is not the final logo and could possible change drastically when our app is fully developed but as of right now it is our current Logo

Asking for visualization methods to get a better understanding of how we could visualize data to our users from a wide range of ages. We discussed and felt from personal opinions that Bar graphs and pie charts as well as word clouds to help gather interviewee data to help structure the path to build our app

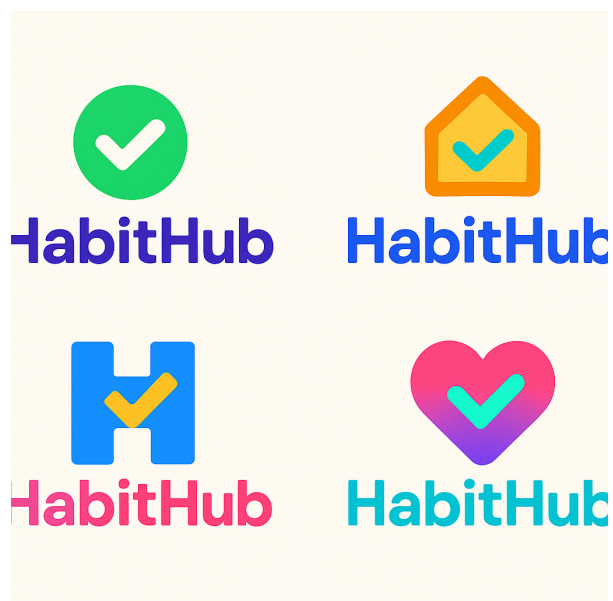
Appendix:

LOGO: I am building an app called habithub a tracking app for people to track their tasks/routines that is gamified. Create 4 logos for me to choose out of

Response:

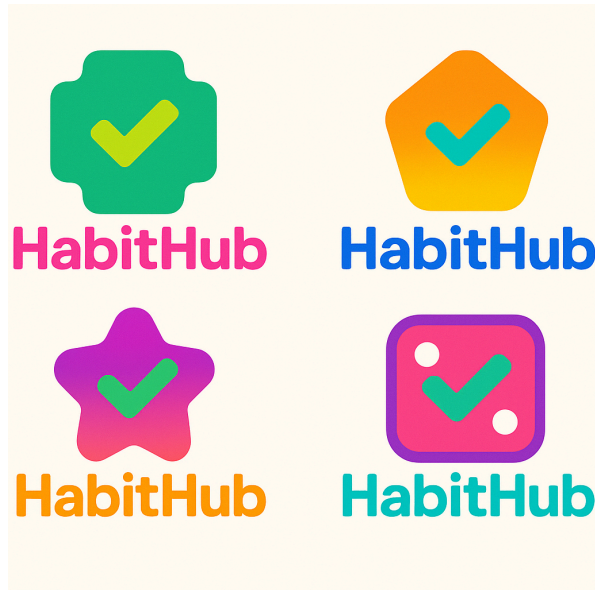


add more colours and also more app driven make it novel and make it colourful and cool looking
the target audience is for kids and parents



Response:

uses different shapes make it look gamified



Response:

gamify it more like its an actual game on the apple store



Response:

lets do bottom right

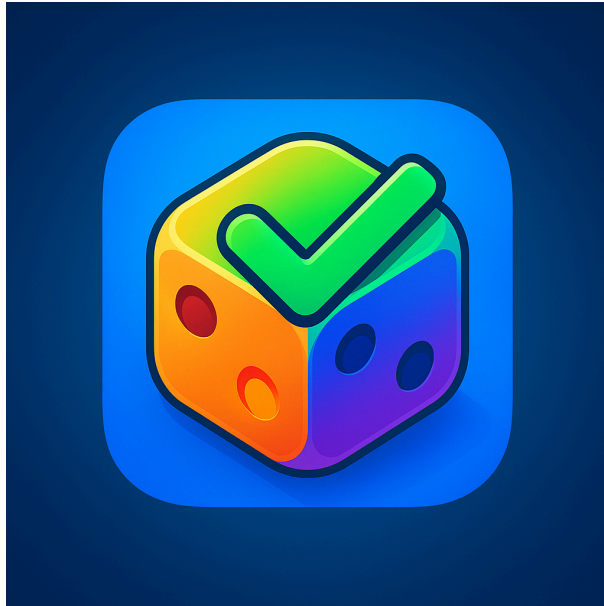
Response: Great choice — the bottom-right logo (dice-shaped icon with checkmark and vibrant colors) captures a playful, gamified feel that's perfect for your target audience of kids and parents.

Would you like me to now create:

- a full app icon version (rounded square format),
- branding mockups (e.g., splash screen, app icon preview),
- or provide logo files and color palettes for usage?

Let me know how you'd like to proceed.

Yes



Response:

VISUALIZATION IDEAS FOR HABITHUB:

Currently I am doing unstructured interviews with children ages (6-11) and their parents for a app I am creating for school. One thing most parents mentioned was to have an easy to read and navigate application. What are some graphs and visualizations I could use to show data. A brief background about the app i am making it is called HabitHub. it is meant to track routines and tasks for children and adults. for a parent they can see their childs progress for what they havent done and have done. what are some graphs and other visualizations I could use. give me a list

Response: Here's a list of **graph and visualization ideas** you can use for your *HabitHub* app, tailored to both **parents** (overview and control) and **children** (fun and gamified visuals):

🧩 For Parents (Data & Progress Visualization)

These should focus on clarity, simplicity, and quick insights.

1. **Progress Bar (Linear or Circular)**

- Shows percentage of daily/weekly habits completed.

- Example: “Emma – 80% of tasks done today.”

2. Bar Graph

- Compare different tasks (e.g., “Homework,” “Brush Teeth,” “Feed Pet”) by completion frequency.
- Great for showing which tasks are consistently missed.

3. Stacked Bar Chart

- Displays completed vs incomplete habits per day/week.
- Helps parents spot trends in their child’s consistency.

4. Line Graph

- Track progress over time (daily streaks or long-term improvement).
- Example: A smooth line showing weekly completion rate over a month.

5. Donut/Pie Chart

- Visual breakdown of task categories (chores, study, personal care).
- Easy to understand at a glance.

6. Heat Map Calendar

- Shows days with full completion in bright colors (green) and missed days in dull tones (gray/red).
- Helps visualize consistency over time.

7. Task Comparison Chart

- Compare performance across multiple children (if a family has more than one kid).
- Example: Simple bar or radar chart.

8. Goal Completion Gauge

- A circular dial showing proximity to a weekly/monthly goal.

- Feels dynamic and intuitive.

For Children (Gamified & Interactive Visuals)

These should be colorful, engaging, and motivating.

1. Character/Avatar Progress Bar

- Character grows, levels up, or gains items as tasks are completed.

2. Star/Badge Collection Board

- A grid showing unlocked achievements or streaks.
- Visually rewards consistency.

3. Animated Progress Meter

- A fun element (like a rocket filling up fuel, or a pet getting happier) when tasks are done.

4. Daily Task Path (Journey Map)

- A trail of steps or icons leading to a goal—each completed task fills a step.

5. Reward Wheel

- Displays possible rewards; spins or lights up when milestones are hit.

6. Color-coded Task Bubbles

- Green = done, Red = missed, Yellow = in progress.
- Can be interactive—kids tap to mark tasks complete.

7. Weekly Streak Tracker

- A simple row of icons (e.g., suns or stars) for each day.
- Encourages streak building.

8. Habit Tree / Garden Visualization

- Each habit represents a plant that grows when the habit is completed.
- Missed days make it wilt slightly—playful but motivating.

Combined Visuals for Family Dashboard

These help bridge the parent-child interaction:

1. **Family Overview Dashboard**

- Mini progress circles for each family member on one screen.
- Example: “Mom: 95%, Dad: 80%, Emma: 60%.”

2. **Task Completion Timeline**

- Shows when habits were completed during the day.
- Could use color-coded blocks (like a simple Gantt chart).

3. **Reward/Level Ladder**

- Everyone’s current level or reward tier side by side.

Would you like me to **mock up a few of these visualizations** (e.g., progress bar, heat map, and child’s badge board) in a colorful, app-style layout (for presentation or Figma design)?

Work Date/Hours

Jayden Prasad

Date	Number of Hours	Description of work done
Sept 24th 2025	2	Initial Brainstorming for project ideas , and written short description of each idea
Sept 30th 2025	1	Created git repo and read me file
Oct 3rd	1	Initial testing and working around of figma seeing what it is and what it offers
October 8th	1	Requirements gathering creating survey questions
Oct 12th	0.5	Interviewed first person
October 13th	1	Interviewed 3 participants for requirements gathering
October 19th	1.5	Played around in the figma basics to get a better understanding on how to build a nice UI
Oct 20th	0.5	Interviewed 2 people for requirements gathering
Oct 21st	1	First attempt at creating home page of Habit Hub
Oct 22nd	1.5	Interviewed 4 participants for requirements gathering
Oct 22nd	0.5	Started working on requirement gathering paper (
Oct 23rd	1	Working on home page for habithub in figma
Oct 24th	0.5	Interviewed 2 participants for requirements gathering
Oct 24th	1	Working on requirements

		gathering paper
Oct 29th	1	Finishing up the home menu prototype for HabitHub in figma
October 30th	1.2	Finishing up requirements gathering paper
Nov 2nd	0.75	Adding final touches to requirements gathering paper to submit

Dhwani Patel

Date	Number of Hours	Description of work done
Sept 24 th	3	Brainstorming and proposing project ideas
Sept 25 th	2	Messed around with Figma and followed a couple short tutorials to get familiarized
Oct 1 st	1	Drafted and prepared project proposal

Oct 5 th	0.5	Finalized proposal and submitted it
Oct 14 th	1.5	Got started on making the wireframe draft for the project app on Figma

Oct 16th	0.5	Interviewed 2 people for the requirements gathering
Oct 20th	1	Interviewed 3 people for the requirements gathering
Oct 22 nd	1	Continued working on the wireframe and picking a color theme and choosing couple aesthetics options